**AGA KHAN UNIVERSITY EXAMINATION BOARD**

**HIGHER SECONDARY SCHOOL CERTIFICATE**

**CLASS XII**

**MODEL EXAMINATION PAPER 2027 AND ONWARDS**

**Physics Paper I**

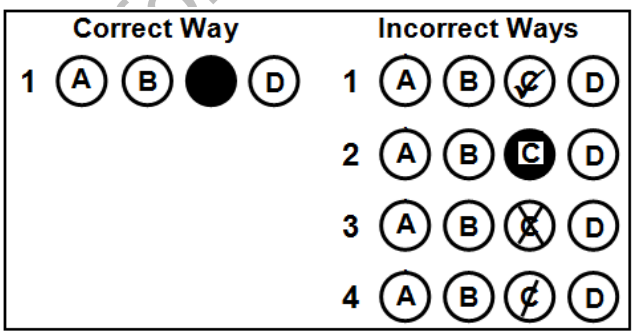
**Time: 1 hour 30 minutes Marks: 50**

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This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

**INSTRUCTIONS:**

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.




**Candidate's Signature**

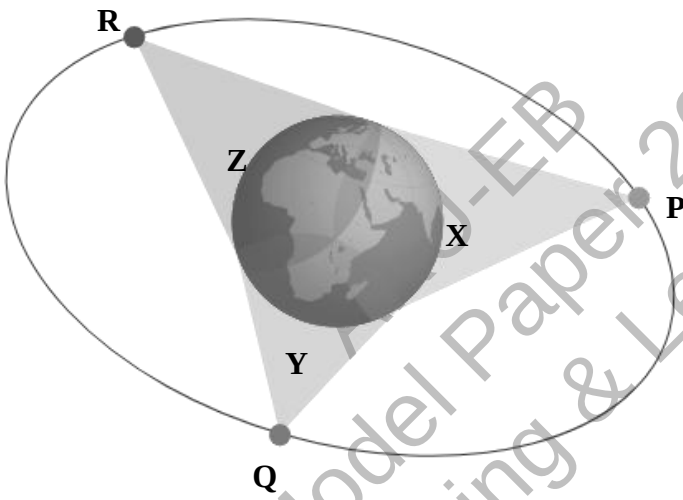
6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. You may use a scientific calculator if you wish.

1. The mass of a planet X is half that of a planet Y, and the radius of the planet X is twice the radius of the planet Y.

   As compared to the planet Y, the acceleration due to gravity on the planet X is

   A. $\frac{1}{8}$ times.
   B. $\frac{1}{4}$ times.
   C. 8 times.
   D. 4 times.

2. In the given diagram **X**, **Y**, and **Z** represent the instantaneous ground distances covered by satellites **P**, **Q**, and **R**, respectively, in a small time interval.




   After 24 hours, the instantaneous distance covered by the satellite **P** on the Earth is

   I. **X**
   II. **Y**
   III. **Z**

   A. I only.
   B. II only.
   C. I and III.
   D. II and III.

3. An electron is placed at each corner of a cube.

   The total charge on that cube is equal to

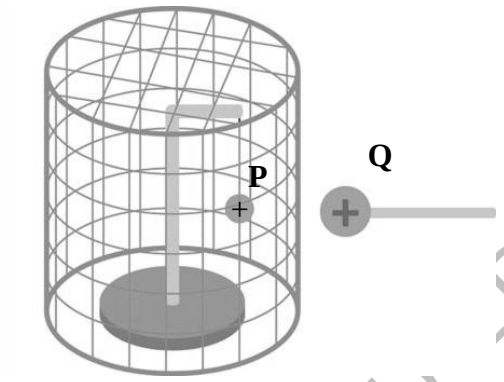
   A. $2\times10^{-20}$ J.
   B. $4\times10^{-20}$ J.
   C. $6.4\times10^{-19}$ J.
   D. $1.28\times10^{-18}$ J.

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4. Coulomb's law is valid in all of the following situations EXCEPT that a charge is placed on a

   A. tree.
   B. wet cloth.
   C. human hair.
   D. stationary wheel.

5. The given diagram shows a metal sphere **P**, containing a positive charge, hanging inside a conducting metal cage.




   If another positively charged sphere **Q** is brought near the outside of the cage, then **P** will

   A. remain at rest.
   B. start vibrating.
   C. move toward **Q**.
   D. move away from **Q**.

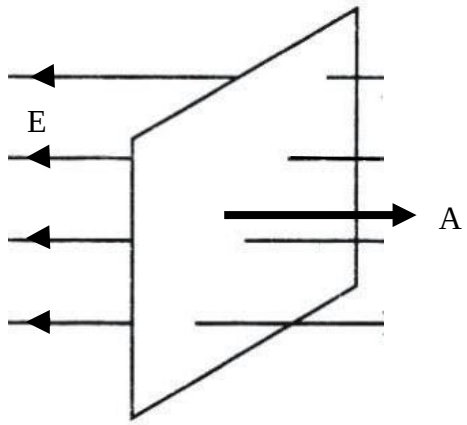
6. A pair of identical particles are placed at two opposite corners of a rectangle.

   Mathematically, the electric field intensity at the centre of the rectangle is

   A. $\frac{kq}{r}$.
   B. $\frac{2kq}{r^2}$.
   C. $\frac{kq_1q_2}{r^2}$.
   D. 0.

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7. The given diagram shows the electric field lines passing through a surface.




The electric flux through this surface is

A. EA.
B. –EA.
C. zero.
D. EAsinθ.

8. If a charge of one Coulomb is moved from one point to another using one joule of energy, then this defines one

A. volt.
B. ohm.
C. tesla.
D. farad.

9. The negative rate of change of electric potential with respect to displacement is

A. capacitance.
B. electric flux.
C. electric field.
D. Coulomb's force.

10. If the air between the parallel plates of a capacitor is replaced by any other dielectric medium, while all other parameters remain the same, then the capacitance will

A. increase.
B. decrease.
C. remain same.
D. become zero.

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11. A wire of cross-sectional area $1.2\times10^{-6}$ m$^2$ carries a current of 1.2 A. If the drift velocity of electrons is 125 m/s, then the number of electrons passing per second through the wire will be

A. 25
B. 126
C. $8\times10^{3}$
D. $5\times10^{22}$

12. The reciprocal of the ability of a material to conduct electric current is known as

A. resistivity.
B. resistance.
C. conductivity.
D. conductance.

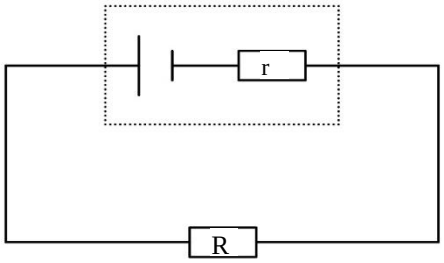
13. By increasing the intensity of light falling on a light dependent resistor (LDR), its resistance will

A. increase.
B. decrease.
C. remain same.
D. become zero.

14. The component whose resistance increases or decreases with an increase in temperature is known as

A. thermistor.
B. tungsten lamp.
C. variable resistor.
D. light dependent resistor.

15. The given diagram shows a source of electromotive force (e.m.f.) connected to an external resistance (R).




If **(**Vt) is the terminal voltage across (R), then the expression for internal resistance (r**)** will be

A. $\frac{\varepsilon - V_t}{I}$.
B. $\frac{\varepsilon + V_t}{I}$.
C. $\frac{V_t - \varepsilon}{I}$.
D. $\frac{V_t}{I}$.

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16. The power loss due to resistance in an electric circuit can be calculated by

A. $\frac{I^2}{R}$.
B. $I^2R^2$.
C. $\frac{V^2}{R}$.
D. $IR^2$.

17. The Kirchhoff ‘s current law (KCL) is based on the law of conservation of

A. heat.
B. charge.
C. energy.
D. momentum.

18. In a Wheatstone bridge, the number of variable resistors used is/ are

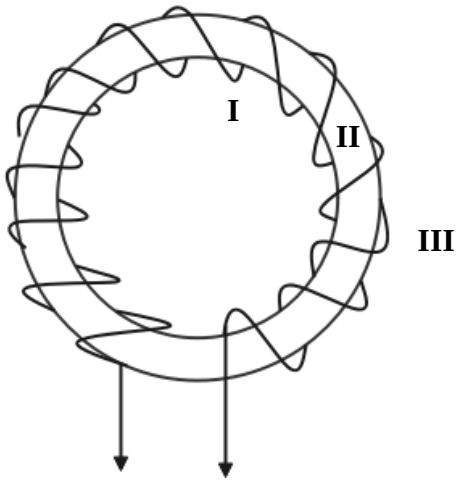
A. 1
B. 2
C. 3
D. 4

19. The magnetic field produced by a long straight current-carrying wire is

A. inversely proportional to the current.
B. directly proportional to the length of the wire.
C. directly proportional to the thickness of the wire.
D. inversely proportional to the distance from the wire.

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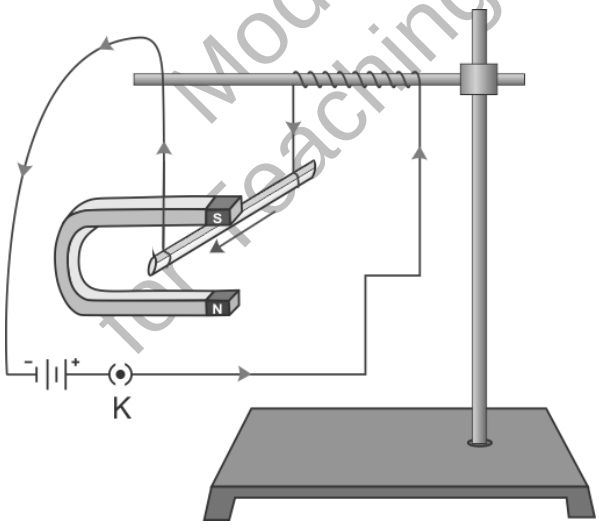
20. The diagram shows a toroid carrying current.




The magnetic field at **I**, **II**, and **III** respectively are

A. $\frac{\mu_o NI}{2\pi r}$, 0, 0
B. 0, $\frac{\mu_o NI}{2\pi r}$, 0
C. 0, 0, $\frac{\mu_o NI}{2\pi r}$
D. $\frac{\mu_o NI}{2\pi r}$, $\frac{\mu_o NI}{2\pi r}$, 0

21. From the given diagram, the direction of the force experience by the rod is towards



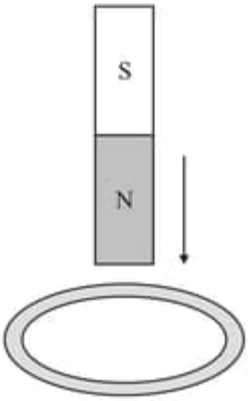

A. up.
B. left.
C. right.
D. down.

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22. A galvanometer can be used as a potential measuring instrument by connecting a

A. high resistance in series.
B. shunt resistance in series.
C. high resistance in parallel.
D. shunt resistance in parallel.

23. The given diagram shows a bar magnet falling through a metal ring.




The force experienced by the magnet while entering and leaving the metal ring would be

| | **Entering** | **Leaving** |
|---|---|---|
| A | attractive | attractive |
| B | repulsive | repulsive |
| C | attractive | repulsive |
| D | repulsive | attractive |

24. In a transformer, the phenomenon of current induced in the primary coil due to an alternating input source is

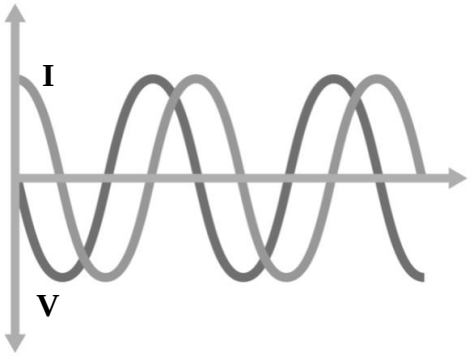
A. self induction.
B. self inductance.
C. mutual induction.
D. mutual inductance.

25. If the peak value of an alternating current is 1.23 A, then the root mean square (rms) value of the current will be

A. 0.615 A.
B. 0.869 A.
C. 1.230 A.
D. 1.739 A.

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26. The following graph shows the variation of alternating voltage and current with time.




Based on the given graph, the CORRECT statement is

A. voltage lead by $90^o$.
B. current lag by $120^o$.
C. voltage lead by $180^o$.
D. current lag by $270^o$.

27. Silicon doped with a pentavalent impurity is known as

I. *N*-type semiconductor
II. Intrinsic semiconductor
III. Extrinsic semiconductor
IV. *P*-type semiconductor

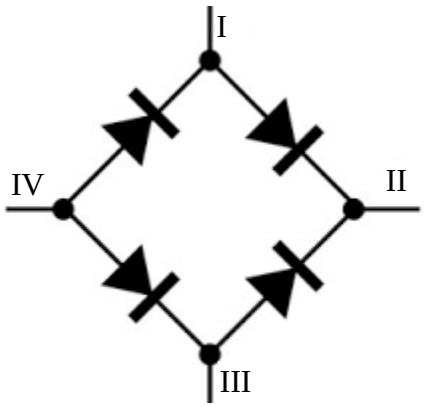
A. I only.
B. II only.
C. I and III.
D. II and IV.

28. When the *p*-type region of p-n junction is connected to the low potential of a cell and the *n*-type region is connected to the high potential of the cell, then the thickness of the depletion region will

A. increase.
B. decrease.
C. become zero.
D. remain the same.

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29. The given diagram shows a part of a bridge rectifier circuit.




The input and output terminals are

| | Input | Output |
|---|---|---|
| A | I and II | III and IV |
| B | II and IV | I and III |
| C | I and III | II and IV |
| D | III and IV | I and II |

30. Suppose you are driving a car at the night time with half of the speed of light.

If you switch on the headlights of the car, then the light will travel with

A. double the speed of light.
B. half of the speed of light.
C. half of the speed of the car.
D. double the speed of the car.

31. Two stationary observers, one on the Earth and the other is on the Moon, measure the length of a rocket moving with a very high speed relative to both the observers.

The measured length of the rocket depends on the

A. mass of the rocket.
B. position of the observer.
C. gravitational field of the Earth and the Moon.
D. relative velocity between observer and rocket.

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32. The light received from distant stars sometimes shows a change in its observed frequency.

This effect occurs due to

A. refraction.
B. diffraction.
C. relative motion.
D. dimension of space.

33. Planck's constant '*h*' has the same unit as

A. angular momentum.
B. kinetic energy.
C. work done.
D. torque.

34. If momentum of a photon X become twice than that of a photon Y, then the ratio of their energies will become

A. 1:2
B. 1:1
C. 2:1
D. 2:2

35. According to the quantum theory of photoelectric effect, if the intensity of incident light increases, then the

A. photoelectric effect will begin.
B. stopping potential will increase.
C. number of emitted electrons will increase.
D. kinetic energy of emitted electrons will increase.

36. If an electron at rest combines with a positron at rest, then the process results in the emission of

A. one photon of energy 0.51 MeV.
B. two photons of energy 0.51 MeV.
C. one photon of energy 0.102 MeV.
D. two photons of energy 0.102 MeV.

37. The mass of a proton is 1.007276 u. The energy equivalent of this mass is approximately

A. 9.38 MeV.
B. 93.82 MeV.
C. 938.27 MeV.
D. 9382 MeV.

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38. The decay constant of a radioactive substance is $1\times10^{-4}\ s^{-1}$. If 500 atoms remains in a sample in 7200 s, then the initial number of atoms present was approximately

A. 243.37
B. 500.50
C. 999.85
D. 1027.21

39. If $X_o$ is the initial number of atoms of a radioactive sample, then the number of atoms remaining after four half-lives will be

A. $\frac{X_o}{2}$.
B. $\frac{X_o}{4}$.
C. $\frac{X_o}{8}$.
D. $\frac{X_o}{16}$.

40. All of the following are the types (flavours) of quarks EXCEPT

A. strange.
B. down.
C. out.
D. top.

41. In modern particle physics, all the matter are classified into two broad categories based on their intrinsic spin.

These categories are known as

A. quarks and leptons.
B. hadrons and leptons.
C. baryons and mesons.
D. fermions and bosons.

42. The ability of certain materials to produce an electric potential when subjected to mechanical vibration is referred as

A. piezoelectric effect.
B. photoelectric effect.
C. thermoelectric effect.
D. electromagnetic induction.

43. If two media have acoustic impedance $4\times10^6$ and $2\times10^6$ respectively, then the ratio of incident and reflected intensities is

A. 1:3
B. 1:9
C. 3:1
D. 9:1

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44. The wavelength of maximum intensity in the spectrum of the Sun is approximately $5\times10^{-7}$ m.

Based on the given value, the surface temperature of the Sun is

(**Note:** Wien's displacement constant value is $2.9\times10^{-3}$ mK.)

A. $5.8\times10^{9}$ K.
B. $5.8\times10^{3}$ K.
C. $1.72\times10^{-4}$ K.
D. $1.45\times10^{-9}$ K.

45. In a medical ultrasound, the image is produced when ultrasound waves

A. produce ionisation inside the body.
B. undergo diffraction causing acoustic impedances.
C. are refracted differently by tissues of different densities.
D. are reflected between tissues having different acoustic impedances.

46. The statement that is CORRECT about X-rays is that they

A. are longitudinal in nature.
B. have wavelengths greater than $10^{2}$ m.
C. fall in visible range of electromagnetic wave.
D. are not deflected by electric and magnetic fields.

47. A star can be used as a standard candle for measuring astronomical distances only if its

A. intrinsic luminosity is known.
B. apparent brightness remains the same.
C. spectrum lies entirely in the visible region.
D. observed intensity increases linearly with distance.

48. The redshift observed in the light from distant galaxies increases with their distance from the Earth.

This observation is best explained by assuming that

A. distant galaxies emit light of longer wavelengths.
B. galaxies are moving through a stationary universe.
C. interstellar dust absorbs shorter wavelengths of light.
D. expansion of universe is stretching the wavelength of light.

49. One argument in favour of continuing research into subatomic particles, despite the existence of nuclear weapons, is that such research

A. helps in development of human clone.
B. can provide evidence for alien's existence in the universe.
C. prevents the misuse of scientific knowledge by governments.
D. contributes to scientific progress and beneficial technologies.

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50. A boy is standing in an elevator trying to measure the mass of an object hanged in a spring balance inside a free falling elevator.

The given situation is an example of a/ an

A. thought experiment.
B. observational study.
C. computer simulation.
D. laboratory experiment.

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